Schema2Bench: Evidence-Grounded Cross-Domain Text-to-SQL Benchmark Migration

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***Abstract*—Text-to-SQL benchmarks compare models that answer natural-language questions over relational databases. Reusing their query structures can extend evaluation to new databases without constructing every task from scratch. However, executable SQL can misrepresent the intended task, and changes in workload composition can alter model comparisons. Structural coverage alone does not establish whether a migrated benchmark preserves source behavior or reflects model differences on the target database. In this paper, we propose Schema2Bench (S2B), a framework for cross-domain benchmark migration. Benchmark Construction grounds source query patterns in target schemas and sampled values, then generates questions from SQL. Workload Selection adjusts composition under fixed quotas, bounds fidelity changes and identifies structural deficits in the candidate pool. Benchmark Evaluation separately measures source fidelity and agreement with model accuracy gaps on target references. We compare five construction methods on eight BIRD databases with ten models, 2,000 tasks per method and 906 target references. In the main comparison under Verified execution scoring and declared weights, S2B reduces the template baseline's target gap from 0.225 to 0.169 and improves source fidelity over all four comparators. Fixed-SQL question rewriting lowers target gap but worsens source fidelity, showing why benchmark continuity and target model selection require separate assessment.**

I. Introduction

Text-to-SQL systems let users query relational databases through natural language. Recent methods improve SQL generation through prompting, specialized models and feedback [1–3]. Their evaluation relies on workloads such as Spider and BIRD [4, 5], whose questions refer to particular tables, relationships and values. Choosing a model for another database requires tasks grounded in that database. Reusing source query structures can reduce construction effort, but the resulting workload must also be assessed as a measurement instrument.

Existing approaches address different parts of this problem. RingSQL uses schema-independent templates to produce training pairs, while OmniSQL synthesizes diverse databases, queries and questions for training [6, 7]. SQLMorph expands joins and perturbs questions to diagnose model weaknesses [8]. These methods establish useful construction and diagnostic techniques. Their generation quality, structural coverage or downstream training gains do not, however, determine whether a migrated workload preserves source model comparisons or supports model selection on the target database. We study this distinction in *cross-domain benchmark migration*.

Challenges. *First, executable SQL does not establish task meaning.* Execution-based evaluation can miss semantic differences [9], and data-grounded explanations show why database contents matter when checking SQL [3]. When transferring a query pattern, matching column types alone can change the entities, relationships or measures being queried. Value-linking studies further show that syntactically valid queries can use the wrong database values [10]. Benchmark construction therefore needs to ground both SQL and its question in the target schema and data.

An illustrative example. Figure 1 asks which customers placed at least two orders. `COUNT(*) >= 2` works on source order rows. On target order lines, C1 has two rows for one order; C2 has two orders. The transferred query returns both; `COUNT(DISTINCT oid)` correctly retains C2. Execution alone cannot distinguish counting units.

Second, a valid workload can change model comparisons. Diagnostic perturbations expose sensitivity to query structure and wording [8, 11]; benchmark-design and concurrence studies also distinguish evaluation claims from the evidence needed to support them [12, 13]. Matching structural frequencies alone need not preserve model behavior. Selection can adjust frequencies only within the available candidate pool, and source fidelity need not imply agreement with target references.

Suppose models A and B score 75% and 50% on the source, but 50% and 75% on target references. Preserving source behavior favors A; target alignment favors B. Compact evaluation methods estimate performance on an existing benchmark [14]; migration changes both the database and task population.

Our methodology. Schema2Bench (S2B) has three modules (Fig. 2). *Benchmark Construction* instantiates source query patterns with target schemas and sampled values, then generates questions from SQL. *Workload Selection* adjusts composition under fixed quotas, tightly bounds fidelity changes and identifies coverage deficits that selection cannot remove. *Benchmark Evaluation* separates source fidelity from target alignment and retains unknowns in direct comparisons. Screening uses earlier outputs; selection reads no evaluator responses. Execution and consistency checks screen operational failures without certifying semantics.

Experimental evaluation. On eight BIRD databases with ten models, S2B improves both source fidelity and target gap

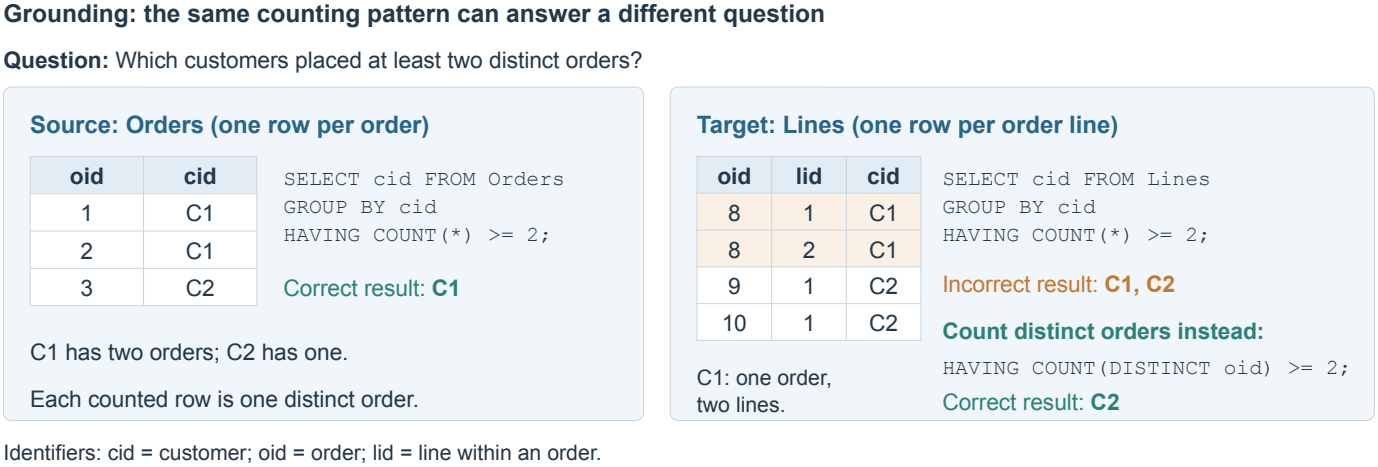


Fig. 1. An illustrative grounding error: transferring `COUNT(*)` from orders to order lines changes what is counted. Both target queries execute, but only distinct-order counting answers the question. Data are constructed for explanation.

over Template. It also improves Verified source fidelity over three generation-based adaptations, which attain lower target gaps. Ablations locate most improvement before selection and show how wording changes this tradeoff.

Contributions. We make three contributions:

1) **Benchmark Construction.** We construct target tasks from source SQL patterns using schema and value evidence, with explicit question–SQL alignment and failure handling (Section IV-A).
2) **Workload Selection.** We bound the fidelity change from quota-constrained selection and characterize structural deficits in the candidate pool (Section IV-B).
3) **Benchmark Evaluation.** We distinguish source fidelity from target model choice and evaluate construction, selection and wording effects under execution and semantic uncertainty (Sections IV-C and V).

II. Preliminaries

A. Text-to-SQL Tasks

A domain is a fixed database snapshot. Task (d_i, x_i, s_i) pairs question x_i and reference SQL s_i on domain d_i. Migration (Fig. 2) constructs N tasks across D domains, using source queries $\mathcal{S}$ for patterns and statistics. A separate workload $\mathcal{W}_s$ provides the behavioral reference for $M \geq 2$ models.

B. Evaluation Objectives

Source fidelity measures continuity with source behavior; target alignment, rankings and regret assess target model comparisons and selection (Section IV-C1).

III. Schema2Bench Overview

In this section, we first formulate benchmark migration in Section III-A, then outline Schema2Bench in Section III-B. Its three modules are *Benchmark Construction*, *Workload Selection* and *Benchmark Evaluation*. Sections IV-A, IV-B and IV-C describe their implementations, respectively.

A. Problem Formulation

Candidate pool $\mathcal{C}$ retains IDs and evidence. Membership $z \in \{0,1\}^{|\mathcal{C}|}$ specifies release $\mathcal{W}_z$, selecting item i when $z_i = 1$. Strata partition candidates within domains; $A_{g,i}$ indicates membership in stratum g, with quota q_g. Feasible memberships $\mathcal{F}$ satisfy $Az = q$ and $\mathbf{1}^\top z = N$, where $\mathbf{1}$ is the all-ones vector. Quotas fix domain sizes $N_d > 0$ and require sufficient candidates per stratum.

Given initial release $z^0 \in \mathcal{F}$, selection seeks

$$\min_{z \in \mathcal{F}} L_{\text{struct}}(z) \quad \text{subject to } H(z, z^0) \leq \tau. \tag{1}$$

L_{struct} measures source-structure mismatch. H bounds changes in source fidelity error (TEF), with tolerance τ (Section IV-B), not absolute loss.

B. Framework Overview

Figure 2 connects candidate construction, workload selection and model evaluation on source, selected and reference tasks.

IV. Methodologies

We detail the three modules below.

A. Benchmark Construction

Schema2Bench instantiates source operator patterns with target tables, types, keys and sampled values, then derives questions from SQL. The evaluated implementation is programmatic; Section V-C2 tests LLM question generation.

Binding operators to the database. Columns whose names suggest identifiers are excluded from measures and descriptive projections. Text and descriptive-name cues guide projections; categorical cues guide grouping. SUM requires cues that the column represents an additive measure. Join candidates follow declared keys or entity-key heuristics. Equality predicates use observed values, with up to fifty samples per column. If a join cannot be built, the generator falls back to a grounded filter or scalar aggregate. Structural features are recomputed from

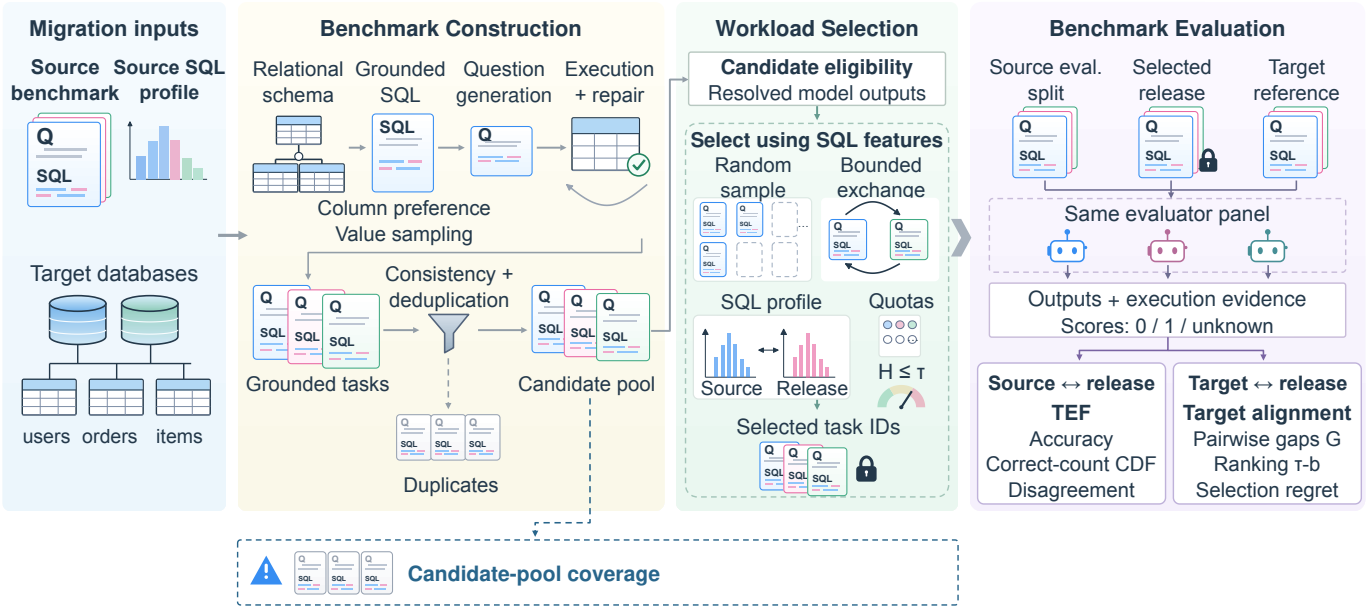


Fig. 2. The three Schema2Bench modules: Benchmark Construction, Workload Selection and Benchmark Evaluation. Q: question; CDF: cumulative distribution function; H: fidelity-change bound; τ: tolerance; G: target gap; τ-b: Kendall rank correlation.

the final SQL; the recorded construction stratum is retained separately.

Keeping the question aligned with SQL. Questions name columns, filters, join relationships and aggregation roles. For recognized joins, `COUNT(*)` describes matched rows: order-line rows in Fig. 1. S2B does not infer a general entity-preserving `COUNT(DISTINCT)` repair; distinct-order counting requires different SQL. Relationship clarification also applies to Template.

Admission and failure handling. Execution checks and bounded repair handle operational failures. A final LIMIT clause requires recoverable ordering over projected columns or proof of a unique cutoff; the question must state the ordering and limit. Consistency checks screen literal, entity and ordering mismatches; deduplication removes repeated tasks. These checks reduce identifiable failure modes without certifying question–SQL meaning. Canonical IDs pair Template and S2B tasks without assuming identical semantics (Section V-D2).

B. Workload Selection

Candidate eligibility. Definite execution scores and normal termination determine $\mathcal{C}$ (Section V-A). This screen depends on outputs; selection reads only SQL features, schema evidence, metadata and a seed. Main-study strata combine domain and recorded construction group.

Selection using SQL features. Sampling q_g candidates per stratum gives initial release z^0. Exchanges improve structural matching while limiting their possible effect on the measured response profile. They target Eq. (1) while preserving quotas. Structural loss averages operator-event and SQL-only difficulty total variation (TV), six shape deviations, relative mean join-count and SQL-length errors:

$$L_{\text{struct}}(z) = \frac{1}{5}\left(\frac{\|\widehat{P}_o - P_o^*\|_1}{2} + \frac{\|\widehat{P}_d - P_d^*\|_1}{2} + \frac{\|\widehat{P}_h - P_h^*\|_1}{6} + \frac{|\widehat{j} - j^*|}{j^*} + \frac{|\widehat{\ell} - \ell^*|}{\ell^*} \right). \tag{2}$$

P_o, P_d describe operator-event and difficulty distributions; P_h contains rates of filtering, joins, grouping, aggregation, ordering and LIMIT. j, ℓ denote mean joins and SQL length in whitespace-separated tokens. Hats and stars denote release and source statistics, respectively; $\|\cdot\|_1$ sums absolute coordinates. Difficulty and shapes count tasks. Positive source means allow relative errors above one.

SQL-only difficulty sums joins, twice the additional SELECT and set-operator counts (UNION, INTERSECT, EXCEPT), and indicators for grouping, HAVING, aggregation, ordering, CASE, DISTINCT and at least two AND or OR tokens. Scores ≤ 1, 2–3 and ≥ 4 define easy, medium and hard, differing from Spider labels and behavioral or semantic difficulty.

Let r_d count tasks removed from initial release z^0 in domain d. The budget function $H = H(z, z^0)$ is

$$H = \sum_d (w_d^A + w_d^R)\frac{r_d}{N_d} = \frac{r}{N} + \frac{1}{D}\sum_d \frac{r_d}{N_d}, \qquad r = \sum_d r_d, \tag{3}$$

where $w_d^A = 1/D$ weights accuracy and $w_d^R = N_d/N$ weights the correct-count cumulative distribution function (CDF) and disagreement across domains (Section IV-C1). Each exchange e replaces an initial item with an unused candidate outside the initial release in the same stratum. For resulting membership z', gain is $\Delta_e = L_{\text{struct}}(z) - L_{\text{struct}}(z')$ and cost is $c_d =$

$1/N + 1/(DN_d)$. Choose the largest positive Δ_e/c_d fitting the budget, breaking ties by resulting loss and IDs. Update z, H; stop at no admissible improvement or the step cap. Feasibility and decreasing structural loss hold, without global optimality.

For fixed source moments, joint response laws and domain counts, $|L_{\mathrm{TEF}}(z) - L_{\mathrm{TEF}}(z^0)| \le H$. Response vectors at Hamming distance h_H change accuracy by at most h_H/M, and CDF plus half-weighted disagreement by at most $\frac{h_H}{M} + \frac{h_H(M-h_H)}{M(M-1)} \le 1$. Coupling removed and added items gives Eq. (3), attained by all-zero source and initial responses with all-one replacements. This perturbation bound also holds in expectation under fixed joint laws, including deterministic saved responses.

1) Candidate Coverage: Over SQL types T, such as easy, medium and hard queries, let p and q_z denote source and release proportions, distinct from quotas q. The convex hull $\mathcal{Q} = \mathrm{conv}\{q_z : z \in \mathcal{F}\}$ contains their feasible mixtures v. With dual type weights h, the minimum type mismatch is

$$\delta_{\mathrm{sup}} = \min_{v \in \mathcal{Q}} \mathrm{TV}(p, v) = \max_{h \in [0,1]^{|T|}} \left(h^\top p - \max_{z \in \mathcal{F}} h^\top q_z \right). \quad (4)$$

The coverage lower bound δ_{sup} uses a bounded-count linear program (LP) relaxation and sorted dual weights. Omitting $H \le \tau$ preserves validity for budgeted selection. An attaining integer release certifies optimal type matching; otherwise integrality may leave a gap. With unchanged feasible quotas, adding b eligible candidates reduces TV over pooled tasks by at most b/N: replace new selections with unused old items in the same strata. This bounds structural capacity, not TEF (Section V-C4); Section V-C5 tests acquisition.

C. Benchmark Evaluation

The same M models answer source, release and reference tasks; unresolved scores remain unknown.

1) Evaluation Measures: **Source fidelity (TEF).** $Y_i \in \{0,1\}^M$ contains task i's execution-match indicators, with m, n indexing models. TEF compares accuracy, correct-count CDF indicators $\mathbf{1}\{\sum_m Y_{i,m} \le k\}$, $k = 0, \ldots, M-1$, and disagreement $|Y_{i,m} - Y_{i,n}|$. Here $\mathbf{1}\{\cdot\}$ is one when true, zero otherwise. With source and release feature means Ψ_s, Ψ_z,

$$L_{\mathrm{TEF}}(z) = \sum_\nu w_\nu |\Psi_{z,\nu} - \Psi_{s,\nu}| = D_{\mathrm{diff}} + D_{\mathrm{acc}} + \tfrac{1}{2} D_{\mathrm{dis}}. \quad (5)$$

ν indexes coordinates with weights w_ν: $1/M$ for accuracy and CDF and $1/(2\binom{M}{2})$ for disagreement. $D_{\mathrm{diff}}, D_{\mathrm{acc}}, D_{\mathrm{dis}}$ are mean absolute differences in CDF, accuracy and disagreement, respectively. For selected domain items $z_d = \{i : z_i = 1, d_i = d\}$, accuracy coordinate $\Psi^A_{z,m}$ averages databases equally:

$$\Psi^A_{z,m} = \frac{1}{D} \sum_d \frac{1}{N_d} \sum_{i \in z_d} Y_{i,m}. \quad (6)$$

CDF and disagreement pool tasks; source accuracy averages 20 databases. Equal target quotas do not remove unequal source sizes. Block weights $(1, 1, 1/2)$ define the convention; Section V-D4 varies these priorities over the nonnegative weight region without changing the structural objective.

Target alignment. For construction c, let $a^r_{d,m}$ and $a^c_{d,m}$ be model m's execution-match rates on reference and constructed tasks in domain d. Target gap measures differences between their pairwise accuracy gaps:

$$G(c) = \frac{1}{D} \sum_d \frac{1}{\binom{M}{2}} \sum_{m<n} \left| (a^r_{d,m} - a^r_{d,n}) - (a^c_{d,m} - a^c_{d,n}) \right|. \quad (7)$$

Lower TEF and gap are better. For constructions a, b, $\Delta G(a,b) = G(a) - G(b)$ and ΔTEF use first-minus-second losses; negative favors the first. $I_G = G(\mathrm{Template}) - G(\mathrm{S2B})$ reports positive improvement. Multiplying by 100 gives gap percentage points (pp), distinct from accuracy changes. Domain-mean Kendall τ-b measures ranking with ties; regret is target accuracy lost by choosing a release winner. Common accuracy shifts preserve gaps, while near ties can reverse rankings.

2) Execution Scoring: Base scoring compares result values and column counts, including empty-result widths. Non-stop or unresolved outputs receive $[0, 1]$; parser rejection or unordered LIMIT alone does not prove error. Neither scoring layer certifies question–SQL meaning.

The Verified layer resolves unknowns using SQLite errors, row counts, unique ordering and checks of possible LIMIT results tied to original outputs (Table II).

3) Comparisons with Unknown Scores: Unknown scores induce feature intervals. Direct contrasts share source and reference uncertainty; target outcomes remain separate unless prompts and outputs match (Section V-C1). Coordinate relaxation gives conservative outer bounds, not confidence intervals or necessarily attainable endpoints. We never subtract independently optimized row bounds.

Bootstrap intervals condition on stated populations, weights and units; shared models and pools make pairs and releases dependent.

Lower and upper score bounds $[l_{i,m}, u_{i,m}]$ are $[0,0]$, $[1,1]$ or $[0,1]$ for mismatch, match or unknown. Weighted means bound accuracy; CDF contributions lie between $\mathbf{1}\{\sum_m u_{i,m} \le k\}$ and $\mathbf{1}\{\sum_m l_{i,m} \le k\}$. Disagreement is fixed when both scores are fixed, otherwise relaxed to $[0,1]$.

For method-feature intervals $A = [a_-, a_+]$, $B = [b_-, b_+]$, the minimum and maximum absolute distances are

$$d_-(A,B) = \max(a_- - b_+, b_- - a_+, 0), \quad (8)$$

$$d_+(A,B) = \max(|a_- - b_+|, |a_+ - b_-|). \quad (9)$$

For shared source-feature interval S, the discrepancy difference (first minus second) has lower and upper bounds

$$\underline{\Delta} = \min_{s \in S} \{d_-(\{s\}, A) - d_+(\{s\}, B)\}, \quad (10)$$

$$\overline{\Delta} = \max_{s \in S} \{d_+(\{s\}, A) - d_-(\{s\}, B)\}. \quad (11)$$

Both terms share source value $s \in S$. We sum weighted coordinate bounds, evaluating endpoints and piecewise-linear breakpoints. Gap contrasts similarly share signed target-reference model gaps.

TABLE I
AVAILABLE → RETAINED CANDIDATES AND REFERENCES AFTER TEN-MODEL ELIGIBILITY SCREENING. EACH RELEASE SELECTS 250 PAIRS PER DOMAIN.

Domain	Candidate pairs	Reference items
California schools	703→703	85→55
Card games	460→453	171→154
Codebase community	373→356	176→153
European football	613→601	123→91
Financial	383→382	101→82
Formula 1	484→478	172→127
Superhero	508→501	129→118
Thrombosis prediction	311→304	161→126
Total	3,835→3,778	1,118→906

V. EXPERIMENTAL RESULTS

We compare source fidelity across methods (Section V-B), the contributions of construction and selection (Section V-C), and robustness across assumptions and domains (Sections V-D–V-E). TEF assesses source continuity; gap, ranking and regret test whether this also supports target model choice.

A. Experimental Setup

1) Datasets and Models: The primary comparison uses eight BIRD databases (Table I), ten models, 2,000 tasks per method and 906 shared references. Spider supplies 7,000 training queries for source profiles and 1,034 development tasks across 20 databases for evaluation. Prompts include data definition language (DDL) schemas from retained snapshots.

The panel comprises Qwen3-4B (Instruct-2507), Granite-3.3-8B-Instruct [15], Qwen2.5-Coder-7B-Instruct [16], Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct [17], Mistral-7B-Instruct-v0.3 [18, 19], Qwen2.5-14B-Instruct, Qwen3-8B, Gemma-2-9B-it [20] and Phi-4 [21]. Qwen2.5-7B also participates in generation [22].

a) Initial qualification result: The original workload improved in five of eight domains, below the preregistered six-domain requirement. Later Qwen3-8B non-thinking evaluations [23] and scoring changes do not revise this failure.

b) Candidate screening and selection: Both tasks in each pair require normal output termination and definite ten-model scores, using compilation and row-count evidence. Incorrect responses qualify; unresolved or truncated responses do not. Most exclusions involve timeouts or unprojected ordering columns (Table I). Reasons can overlap and exclude earlier generation failures.

We sample 250 pairs per domain within strata from the 3,778-pair pool using proportional largest-remainder quotas. S2B features determine both arms' membership before cached scores are read; Template is not independently optimized. Equal quotas and $H \leq 0.01$ allow ten equal-cost swaps, preventing isolation of normalization benefits. Ten releases overlap; scoring layers share tasks. Outcome-dependent eligibility and unequal precision remain limitations.

2) Baselines: Template tests structural reuse without S2B's grounding; both share the paired pool (Section III). Three external adaptations test generation workflows as benchmark constructors. Ring combines template SQL with Qwen-generated questions. Omni reuses SQL and question generation components with local Qwen models. Synth uses the author's synthesis, question generation, backtranslation and self-comparison workflow with local Qwen, removing its 25-row cap.[1] These adaptations do not reproduce the complete original systems. Evaluation sizes are fixed; generation budgets differ (Table VII).

Each external method supplies one unpaired set of 2,000 tasks. Omni replaces duplicates before response inspection; Synth expands adaptively and deduplicates in fixed order. Synth combines reused and new outputs with different batch and key-value cache settings; interrupted shards leave 366 tasks without post-run model checks.

3) Scoring Settings: Both scoring layers use the same tasks. Verified scoring resolves unknowns using saved execution evidence, with unequal coverage concentrated in California for the first four methods. Detailed score counts accompany the artifacts; semantic assessment is separate (Section V-D2).

B. Main Results

1) Source Fidelity and Target Alignment: Schema2Bench improves both objectives over Template: lower source TEF and a **0.055693** reduction in Verified target gap (Table II). All four direct TEF bounds are strictly negative, establishing a fidelity advantage over the three generation-based adaptations despite overlapping marginal intervals. Ring, Omni and Synth attain lower gaps. S2B therefore offers better source continuity, while target alignment favors those alternatives.

All four mean Verified TEF contrasts remain negative across ten overlapping releases using fixed external sets, testing membership sensitivity rather than independent construction. Base mean bounds cross zero for Template, Ring and Synth; only the Omni contrast remains negative. The advantage over all four methods depends on additional execution evidence.

2) Model Ranking: Average model ordering also improves over Template: across ten Verified releases, mean gap falls from 0.224640 to 0.168988 (**5.5652 pp**) and mean domain τ-b rises from 0.2292 to 0.3272. California and codebase lose rank agreement, limiting the domain-level conclusion.

For each seed, 1,000 bootstrap draws separately resample matched SQL-skeleton clusters and references. All ten rank-difference estimates are positive; eight conditional 95% intervals exclude zero. These intervals omit missingness and pool selection.

3) Model Selection: We test whether fidelity improves model selection using a regret rule fixed before this post-hoc calculation. It retains all possible winners under ties and unknowns, without ranking midpoints.

With 906 references and Verified scoring, mean regret favors Ring despite S2B's lower TEF (Table III). Domain

[1] SynthSQL toolkit: https://github.com/NeginKhalifat/SQL-benchmark, retained commit `dff8c75e`. It is distinct from SING-SQL and SynSQL.

TABLE II

FIRST RELEASE: 2,000 TASKS PER METHOD, TEN MODELS, 906 REFERENCES AND 1,034 SPIDER ITEMS. DIRECT TEF DIFFERENCES SHARE SOURCE UNKNOWNS; NEGATIVE FAVORS S2B. UNKNOWNS ARE OUT OF 20,000 SCORES. BOLD MARKS BEST AND UNDERLINING SECOND WITHIN COMPARABLE GROUPS; UNRESOLVED RANKS, DIRECT CONTRASTS AND DIFFERENT SENSITIVITY SETTINGS ARE UNRANKED.

Method	Target gap $G \downarrow$	Spider–BIRD TEF $\downarrow$	Direct ΔTEF	Unknown
Verified execution scoring				
Template	$[0.224967, 0.224967]$	$[0.243915, 0.390986]$	$[-0.131730, -0.079099]$	0
S2B	$[0.169275, 0.169275]$	$\mathbf{[0.138501, 0.283651]}$	0	0
RingSQL-style adaptation	$[0.096309, 0.100791]$	$[0.283403, 0.435663]$	$[-0.233594, -0.061506]$	46
OmniSQL component adaptation	$[0.029731, 0.134760]$	$[0.564696, 0.862439]$	$[-0.665769, -0.324655]$	1,290
SynthSQL-style local adaptation	$[0.013004, 0.166928]$	$[0.268596, 0.642072]$	$[-0.453605, -0.021084]$	2,040
Base layer: strict execution and output-shape checks				
Template	$[0.178882, 0.262025]$	$[0.205735, 0.426469]$	$[-0.193294, +0.005344]$	430
S2B	$[0.117662, 0.208049]$	$[0.111299, 0.334757]$	0	537
RingSQL-style adaptation	$[0.059199, 0.138379]$	$[0.241426, 0.462380]$	$[-0.293401, +0.034557]$	445
OmniSQL component adaptation	$[0.010465, 0.190895]$	$[0.488896, 0.898945]$	$[-0.734008, -0.195186]$	2,066
SynthSQL-style local adaptation	$[0.000000, 0.345368]$	$[0.103282, 0.861883]$	$[-0.686921, +0.198272]$	5,149

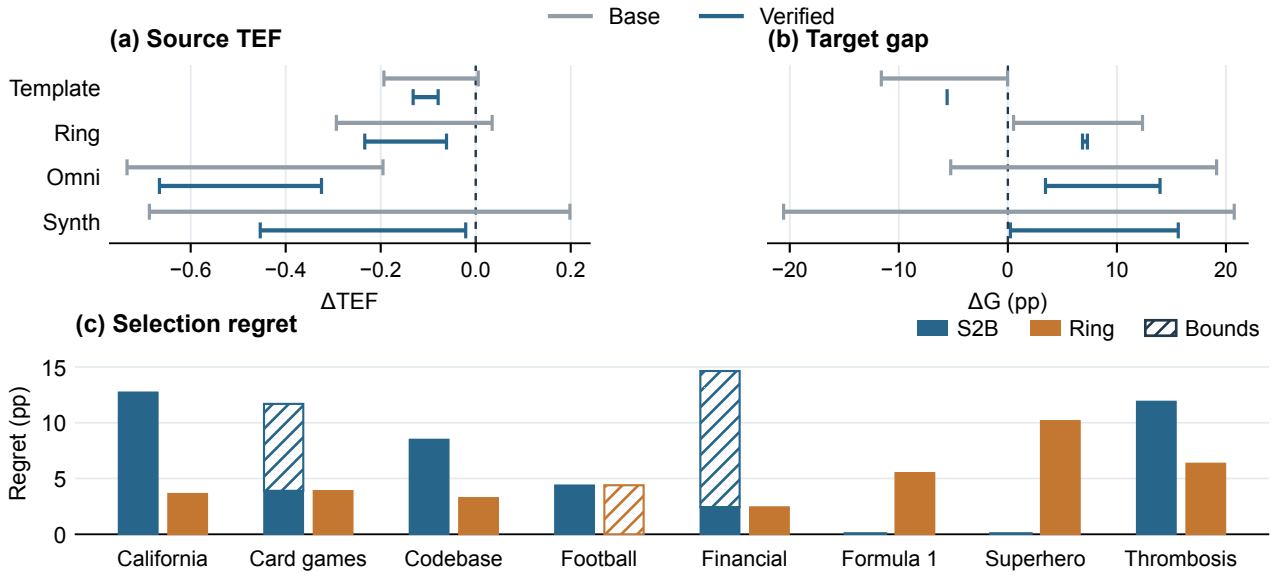


Fig. 3. First-release source TEF, target gap and model-selection regret. Panels (a–b) compare S2B with each method; panel (c) compares S2B and Ring by domain. Caps indicate bounds; ticks indicate coincident endpoints.

comparisons favor Ring in California and codebase, and S2B in Formula 1 and superhero. Omni and Synth bounds remain wide. Figure 3(c) distinguishes lower bounds from remaining ranges.

All 1,118 references yield overlapping S2B and Ring bounds, with 1,356 Base and 706 Verified unknowns. Ten-release regret ranges are retained with the artifacts.

C. Ablation Studies

We test grounding choices and question wording before examining selection (Fig. 4). Grounding controls use separate paired releases; rewriting fixes SQL; selection controls share the pool. Their effects do not additively decompose the main improvement.

1) Column Preference and Value Sampling: To test the grounding choices, paired releases disable descriptive-column preference or reduce sampling from fifty values to five. They use the main models, source tasks and references, with 38 strata, 250 tasks per domain in each arm and $H = 0.01$. Attempts share random states and plans; separately deduplicated candidates must qualify in the same final stratum.

The preference control retains identifier filtering, text preference, grounding and admission rules. Four domains require 1,600 extra slots each before freezing; the arms share 1,159 of 2,000 question–SQL pairs. Sampling uses 2,400 slots per domain in each arm: fifty and five values admit 7,131 and 6,149 candidates, respectively, with 5,375 joint admissions and 769 shared release slots. Admission differences remain part of the comparison.

TABLE III

MODEL-SELECTION REGRET IN ACCURACY PERCENTAGE POINTS, INCLUDING ALL POSSIBLE TIED WINNERS. MEAN AND WORST DENOTE DOMAIN AVERAGE AND MAXIMUM; 906 AND 1,118 ARE REFERENCE COUNTS. COLUMNS SEPARATE SCORING LAYERS; BOUNDS ARE CONSERVATIVE.

Method	906: base mean	906: verified mean	906: verified worst	1,118: base mean	1,118: verified mean
Template	[6.138, 8.979]	[6.718, 7.854]	[10.317, 12.727]	[0.556, 19.177]	[2.213, 14.329]
S2B	[4.122, 9.106]	[5.482, 7.981]	[12.727, 14.634]	[0.071, 19.192]	[2.500, 14.417]
Ring	[2.887, 8.100]	[4.409, 4.958]	10.169	[0.484, 18.847]	[1.547, 11.519]
Omni	[0.000, 17.895]	[0.794, 12.246]	[6.349, 19.512]	[0.000, 28.023]	[0.000, 17.857]
Synth	[0.000, 27.258]	[0.000, 19.046]	[0.000, 25.490]	[0.000, 37.369]	[0.000, 23.858]

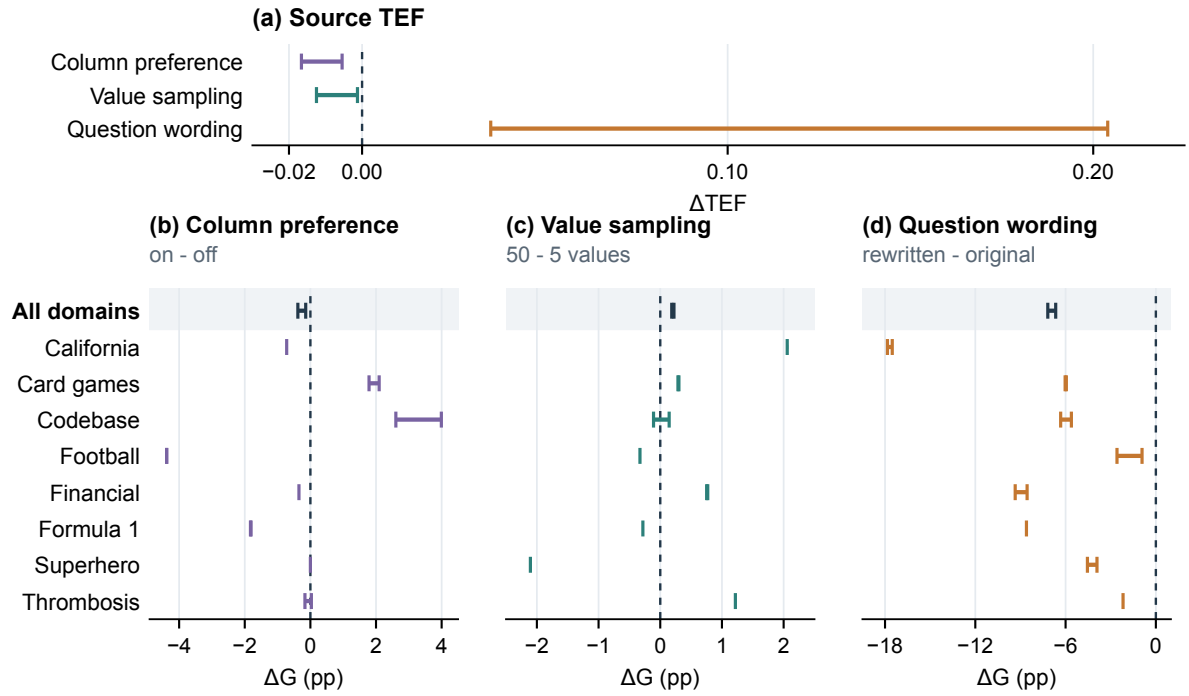


Fig. 4. Construction controls under Verified scoring: source TEF and target-gap differences across domains.

TABLE IV

SELECTION OVER TEN OVERLAPPING 2,000-TASK RELEASES. (A) BIRD, TEN MODELS: POSITIVE I_G FAVORS S2B OVER TEMPLATE. (B) EARLIER SEVEN-DATABASE, SEVEN-MODEL STUDY: LOWER IS BETTER. BEST AND SECOND VALUES ARE MARKED SEPARATELY WITHIN EACH PANEL AND METRIC.

(a) BIRD: construction versus selection			
Selection	Swaps	H	$I_G \uparrow$ (pp)
Initial stratified sample	0	0	5.5386
Matched random swaps	10	0.01	<u>5.5449</u>
Structural adjustment	10	0.01	**5.5652**
(b) Seven additional databases: selection			
Selector	Structural loss		TEF
Stratified random	0.269519		0.050071
Structural greedy	**0.213055**		<u>0.048692</u>
Bounded selection	<u>0.260790</u>		**0.047229**

SQLite compilation resolves 827 preference and 903 sampling-control parser rejections as mismatches without rewriting SQL. Identical requests and outputs share unknowns across arms; others remain relaxed. This post-hoc treatment preserves scores, membership and identity records. Table V also retains earlier independent-arm bounds.

Bounds with shared unknowns support column preference on both metrics and deeper sampling on TEF at a small gap cost (Table V). Grounding benefits therefore depend on the metric. Base TEF favors both, with unresolved gaps.

Column preference increases the gap in card games and codebase (Fig. 4(b)). Excluding European football changes its seven-domain mean difference to $[+0.001899, +0.004625]$; excluding Formula 1 leaves it unresolved. Sampling effects also vary by domain.

a) Value sampling on additional databases: Four SynSQL-2.5M training databases [7] supply 1,000 identical pairs; BIRD `student_club` changes 150 of 250 pairs. Three further targets miss quotas. Across thirteen domains, 3,250 tasks per arm and 1,733 references give Verified TEF and gap differences of $[-0.050927, +0.049902]$

TABLE V
CONSTRUCTION CONTROLS UNDER VERIFIED SCORING. NEGATIVE FAVORS FULL PREFERENCE OR FIFTY VALUES. SHARED VARIABLES BIND IDENTICAL OUTCOMES; INDEPENDENT-ARM BOUNDS RETAIN THE EARLIER RELAXATION.

Intervention	Unknown-variable treatment	Direct TEF difference	Direct target-gap difference
Column preference	Independent arms	$[-0.029752, +0.008062]$	$[-0.011942, +0.006559]$
Column preference	Shared identical requests	$[-0.016593, -0.005466]$	$[-0.003816, -0.001425]$
50 versus 5 values	Independent arms	$[-0.024987, +0.011333]$	$[-0.006878, +0.011031]$
50 versus 5 values	Shared identical requests	$[-0.012477, -0.001261]$	$[+0.001863, +0.002221]$

and $[-0.029026, +0.030559]$, respectively. Adding SQLite evidence only on the original eight domains gives $[-0.013017, +0.011489]$ and $[-0.004886, +0.007287]$. Both remain unresolved. Unchanged pairs, unequal evidence and synthetic training-data overlap limit independent replication.

2) Question Generation: We isolate wording effects by holding SQL fixed. A preliminary comparison identifies 1,973 Ring–Template pairs sharing SQL, domains and schema prefixes but differing in questions. Their Ring-minus-Template execution-match difference is $[-13.96, -13.73]$ pp, motivating this intervention.

We retain all 2,000 first-release S2B queries and quotas, using Ring-style Qwen2.5-7B question generation. The plan and 160-item audit, with 20 items per domain, precede generation. This yields 1,999 correctly formatted questions and 19,990 predictions; the remaining item stays unknown.

Rewriting reduces Verified gap in all eight domains and aggregate Base gap, but increases Verified TEF (Table VI). Thus fixed SQL does not preserve model comparisons when questions change. Concordant pairs increase; regret intervals overlap. Against Ring, TEF is lower but gap remains unresolved.

Using exposed synthetic controls, Terra supports 117 of 160 tasks. Changes in task meaning remain a concern despite fixed SQL (Section V-D3).

3) Workload Selection: Most gap improvement over Template precedes selection (Table IV(a)). Structural adjustment adds only 0.0204 pp over matched random swaps under Verified scoring, improving nine of ten seeds. The 16 alternatives per seed match screening, directed removals, swaps and H. Selection controls composition within the fidelity bound, with little additional gap benefit.

Template and S2B have initial mean τ-b values of 0.2312 and 0.3270, versus 0.2292 and 0.3272 after adjustment. Relative to matched random swaps, TEF changes by $[+0.000227, +0.000977]$ with source summaries (eight seeds unfavorable; two unresolved) and $[-0.000777, +0.000452]$ with source DDL (two favorable; eight unresolved).

a) Seven additional databases: Earlier seven-model results favor bounded selection for TEF and greedy selection for structural loss; all 14 swaps occur in TPC-DS (Table IV(b)). Repair improves yield, but controlled construction produces fewer tasks at higher cost than Ring, without establishing main-study component benefits.

4) Candidate Coverage: Coverage analysis identifies when further selection cannot improve the workload. The pool has 311 easy, 2,688 medium and 779 hard tasks, versus source proportions $(0.464, 0.348, 0.188)$. The 38 quotas allow at most 173 easy items per 2,000; $h = (1, 0, 0)$ certifies $0.464 - 173/2000 = 0.3775$. Both $(173, 1302, 525)$ and the current $(173, 1376, 451)$ attain it. The latter meets $H = 0.01$, certifying that selection cannot reduce difficulty TV. Exact matching needs at least $928 - 173 = 755$ additional eligible easy items, without ensuring joint type feasibility.

5) Additional Candidate Generation: On a SynSQL-2.5M garment database with 159 synthetic references, adaptive expansion follows a 223/250 quota failure. Equal 1,200-attempt supplements yield 119 guided versus 29 untargeted candidates, but all pools retain difficulty-TV optimum 0.356. This bound relaxes H and does not bound TEF.

Ten overlapping releases reuse seven BIRD domains, with 250 tasks per domain, $H \leq 0.01$ and identical structural losses. Guided-minus-untargeted mean TEF is $[-0.000294, +0.000334]$ and new-target gap is $[-0.000905, +0.000542]$. Gap differences are favorable in three releases, adverse in one and unresolved in six. Of 1,590 reference scores, 111 remain unknown. Higher yield leaves these contrasts unresolved and structural deficits unchanged.

D. Robustness Analysis

1) Sensitivity to Sampling and References: For the first-release gap difference -0.0556925, 1,000 bootstrap draws resample paired tasks within domain strata and references within domains, retaining joint model responses. The conditional 95% interval is $[-0.0665464, -0.0434725]$.

Leave-one-domain-out contrasts and all 56 three-domain means remain favorable. These checks condition on observed pools, excluding independent construction variance.

a) A revised target reference: We evaluate all 417 same-domain items in BIRD's October 2026 Mini-Dev SQLite revision.[2] References do not overlap by domain, question and evidence identity. All reference SQL executes; 67 predictions truncate.

Revised-reference gap favors S2B over Template in six domains, with California and Formula 1 unresolved (Table IX); S2B-minus-Ring is $[+0.040718, +0.103446]$. The unpaired revision has more queries rated hard by SQL complexity and more nested SELECTs. Domains and source TEF are unchanged.

[2]https://github.com/bird-bench/mini_dev

TABLE VI

FIRST-RELEASE QUESTION REWRITING WITH FIXED SQL. (A) VERIFIED RESULTS: CONCORDANCES COUNT DEFINITE AGREEMENT AMONG 360 DEPENDENT MODEL PAIRS; REGRET IS IN ACCURACY PERCENTAGE POINTS. GAP AND REGRET RANKS ARE UNRESOLVED. (B) DIRECT DIFFERENCE BOUNDS; NEGATIVE FAVORS REWRITING.

(a) Overall results Release	Target gap ↓	Definite concordances ↑	Mean regret (pp) ↓
Original S2B	0.169275	229/360	$[5.48, 7.98]$
Rewritten S2B	$[0.097452, 0.102706]$	**248/360**	$[4.30, 5.57]$
Ring	$[0.096309, 0.100791]$	<u>246/360</u>	$[4.41, 4.96]$
(b) Direct differences Contrast	Scoring	ΔTEF	ΔG
Rewritten − original S2B	Verified	$[+0.035191, +0.203968]$	$[-0.071823, -0.066569]$
Rewritten − Ring	Verified	$[-0.046174, -0.009796]$	$[-0.003339, +0.006397]$
Rewritten − original S2B	Base	Crosses zero	$[-0.120114, -0.002971]$

TABLE VII

RECORDED TOKENS INCLUDING FAILURES. EVALUATION ROWS ARE RELEASE MEANS; GENERATION ROWS COVER DIFFERENT SCOPES.

Stage	Input tokens	Output tokens	Scope
Template evaluation	22,906,504	1,170,824	20,000 model–item records per release
S2B evaluation	22,936,877	1,143,564	20,000 model–item records per release
Spider summary evaluation	1,816,236	370,097	10,340 source records
Spider DDL evaluation	4,221,154	370,682	10,340 source records
Synth final evaluation	22,041,137	804,975	20,000 records; 5,080 reused
Ring question generation	417,101	125,758	2,400 records; excludes template SQL
Omni SQL and question generation	31,498,785	5,905,818	Initial, additional and reserve generation
Synth all recorded generation	553,507,001	15,411,464	258,674 calls, including failed or stopped attempts

2) Semantic Assessment: OpenAI's `gpt-6-astra` API assesses question–SQL agreement [24, 25] from question, schema and SQL, using a fixed prompt and medium reasoning effort, blind to methods and scores. Qwen and Terra cross-check supported, incorrect and insufficient-evidence labels. Evaluator bias [26] limits these judgments; they are not human verification or calibrated confidence. Sensitivity analyses make flagged scores unknown at fixed denominators.

a) Generated-task audits: The earlier stratified 400-task audit and the sensitivity retaining only 654 nonempty Synth roundtrip matches also leave Synth unresolved; full bounds accompany the artifacts.

The expanded audit fixes 100 records per method in each domain, with one unavailable (Table VIII); exposed controls score 24/24 per class. The union of 1,133 audit concerns and 620 earlier SQL flags makes 1,433 items unknown at the original denominators.

Making all 1,433 flagged items unknown preserves S2B's TEF advantage over Template, $[-0.179682, -0.050176]$, but leaves external contrasts unresolved. Shared-source LP bounds for Ring, Omni and Synth are $[-0.293390, +0.139461]$, $[-0.840707, +0.132174]$ and $[-0.587412, +0.236106]$, respectively; joint gap bounds also cross zero. Three of 48 gap optimizations reach time limits. Outward tolerances and earlier wider bounds remain.

b) Target-reference audits: Parsing JSON inside code fences gives Qwen and Terra agreement on 107 of 160 references (Table VIII(b)); independently generated SQL sometimes contradicts labels. The 105-item strict union treats format failures and non-supported judgments as risks, not established errors.

TABLE VIII

API JUDGMENTS: S, SUPPORTED; I, INCORRECT; U, INSUFFICIENT EVIDENCE; NA, UNAVAILABLE OR INVALID. BOLD AND UNDERLINING RANK SUPPORT COUNTS IN (A) AND (C); (B) COMPARES JUDGES. THESE JUDGMENTS DO NOT VERIFY SEMANTICS.

(a) Astra: 800 tasks per method, 100 per domain Method	S	I	U	NA
Template	**800**	0	0	0
S2B	<u>732</u>	17	51	0
Ring	603	123	73	1
Omni	290	418	92	0
Synth	442	308	50	0
(b) References: 160 tasks, 20 per domain Judge and parsing rule	S	I	U	NA
Terra	89	54	17	0
Qwen: strict JSON	76	9	1	74
Qwen: code-fence parsing	121	37	2	0
(c) Terra: 160 paired tasks, 20 per domain				
Original S2B	**141**	14	5	0
Rewritten S2B	<u>117</u>	32	10	1

The 71-reference scenario favors S2B in all ten releases. Combining generated-task concerns with strict cross-judge

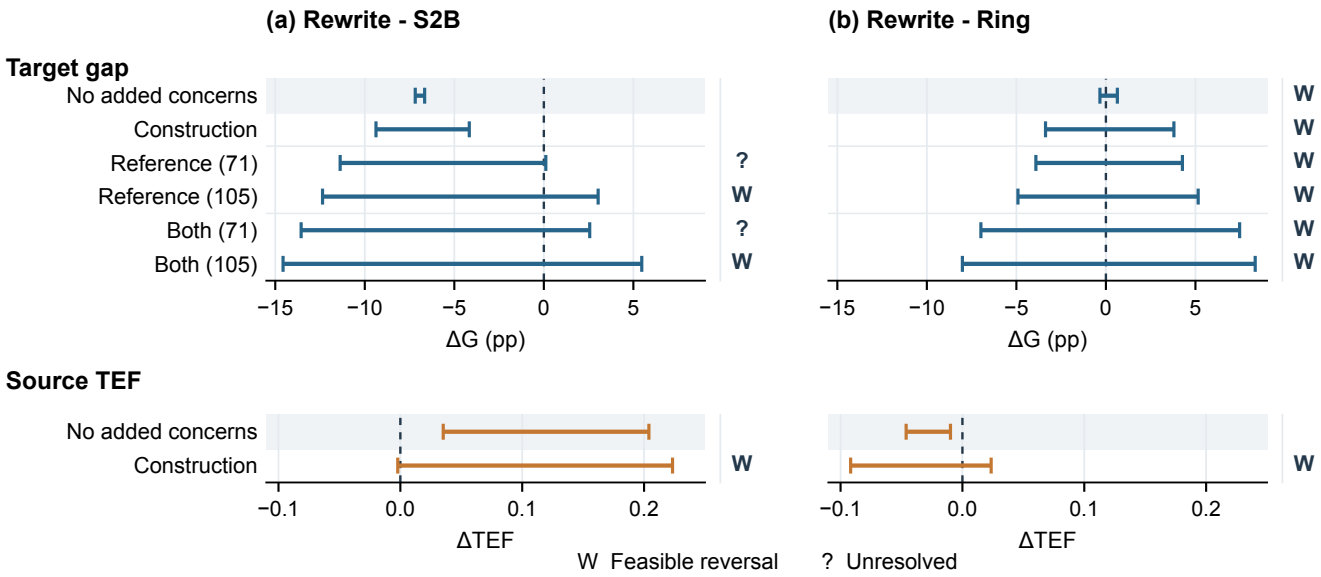


Fig. 5. Sensitivity of rewriting with fixed SQL using earlier audit flags. Target gap is above source TEF. W: a feasible score assignment reverses the comparison; ?: search unresolved; blank: not searched.

TABLE IX
SENSITIVITY OF THE FIRST-RELEASE VERIFIED S2B-MINUS-TEMPLATE GAP. SCENARIOS RETAIN DENOMINATORS AND UNFLAGGED SCORES; INTERVALS BOUND UNKNOWN SCORES.

Generated-task concerns	Reference condition	Reference count	Direct gap difference
None added	Original reference	906	-0.055693
1,433 union flags	Original reference	906	$[-0.077344, -0.026784]$
None added	71 Terra reference concerns	906	$[-0.098374, -0.010170]$
137 original audit flags	105 strict cross-judge concerns	906	$[-0.114491, +0.013951]$
137 original audit flags	79 code-fence sensitivity concerns	906	$[-0.105686, -0.002284]$
None added	Revised BIRD reference	417	$[-0.076995, -0.024912]$
1,433 union flags	Revised BIRD reference	417	$[-0.105428, +0.000213]$

checks or revised references leaves the first release unresolved (Table IX).

3) Sensitivity to Question Rewriting: Figure 5 separates construction concerns (43 rewritten, seven original S2B, 23 Ring) from reference concerns (71 Terra or 105 strict cross-judge flags). Each flag makes ten scores unknown, fixing others and sharing source and reference variables. "Both" combines these flags, excluding the separate 1,433-item union.

Construction flags preserve rewriting's gap advantage but leave both TEF contrasts unresolved. Either reference-risk set makes gap unresolved without changing TEF.

Positive differences are feasible in 14 of 16 Verified cases with bounds crossing zero, without establishing semantic errors or reversal probabilities.

Terra reviews 160 paired originals under the same protocol (Table VIII(c)). Combined flags cover 24 original, 43 rewritten, 23 Ring and 71 reference items. Verified rewrite-minus-original gap $[-0.142130, +0.032135]$ and TEF $[-0.008539, +0.235566]$ remain unresolved.

4) Metric Weights and Model Coverage: Accuracy and correct-count CDF favor S2B over all comparators. Disagreement favors S2B over Template and Ring over S2B ($[+0.012123, +0.066481]$); Omni and Synth remain unresolved. Equal weights and pooled-task averaging leave S2B-minus-Synth unresolved: $[-0.400081, +0.034253]$ and $[-0.430144, +0.025669]$, respectively. Template, Ring and Omni directions persist.

Figure 6 varies nonnegative accuracy, CDF and disagreement weights summing to 2.5 on the first Verified release without semantic flags. Colors classify favorable, adverse and unresolved regions using conservative bounds, not probabilities.

S2B has lower TEF than Template throughout the weight region. Emphasizing disagreement can favor Ring and leaves comparisons with Omni and Synth unresolved.

Removing Qwen and recomputing normalizers preserves favorable Verified TEF contrasts; external adaptations retain lower gap (Table X). Schema-summary and DDL source prompts favor S2B over Template; their direct difference remains unresolved.

Broader coverage preserves S2B's gap advantage over Template (Table X). Adding Qwen3.6-27B, Qwen3.8-27B, Gemma 3 4B and OLMo 2 7B tests model coverage without

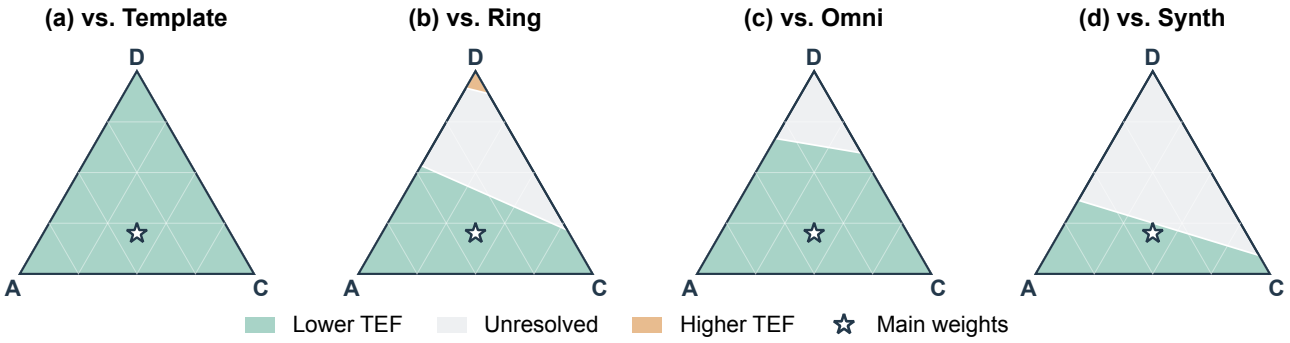


Fig. 6. Post-hoc TEF weight sensitivity. A: accuracy; C: correct-count CDF; D: disagreement. Star: original weights $(1, 1, 0.5)$.

TABLE X
EIGHT-DATABASE GAP SENSITIVITY: POSITIVE $I_G = G(\text{Template}) - G(\text{S2B})$ FAVORS S2B. CONDITIONAL CONFIDENCE INTERVALS (CIS) RESAMPLE WITHIN DOMAINS; BOUNDS DESCRIBE UNKNOWN SCORES.

Setting	Pairs	References	Gap improvement I_G	Interpretation
Non-Qwen panel	2,000	906	0.026560	Ten-release mean
Broader reference coverage	3,778	982	0.053302 $[0.041870, 0.063207]$	Conditional 95% CI
Equal-skeleton weighting	3,778	982	0.046936 $[0.034298, 0.057670]$	Conditional 95% CI
Full available pool	3,835	1,118	$[0.009303, 0.090704]$	Unknown-score bound
Fourteen models: strict	2,000	906	$[0.002662, 0.059226]$	Unknown-score bound
Fourteen models: database checks	2,000	906	$[0.025795, 0.032495]$	Unknown-score bound
Fourteen models: strict	3,778	906	$[0.002188, 0.059640]$	Unknown-score bound
Fourteen models: database checks	3,778	906	$[0.025787, 0.032648]$	Unknown-score bound

The 2,000-pair fourteen-model rows average endpoints over ten overlapping releases.

resolving exclusion bias, transfer or training overlap.

LIMIT appears in 1,739 Template, 2,000 S2B, 1,753 Ring, 424 Omni and 872 Synth tasks, versus 62 of 906 references. Only 108 of 240 method–domain–structure cells, including ten of 40 LIMIT cells, contain ten tasks in both populations, preventing complete matched analysis. A pilot table-name repair does not apply to retained failed outputs.

E. Generalization Experiments

TACO OpenData [27] supplies two related Beijing databases and Connecticut, with potentially synthetic references. Admission and initial sampling differ from the main study. With 16 retrieved objects, contrasts are unresolved against Template and favor Ring (Table XI). Incomplete schemas limit semantic assessment; unfinished Omni and Synth runs miss 250-task quotas.

A ten-model pilot uses CORE-T's SQLite-converted BEAVER [28].[3] DW, Nova and Neutron supply 250, 198 and 24 tasks per method and all 121, 43 and 30 references, respectively. Nova and Neutron share provenance. Retrieval covers required objects for 33 of 194 references. Oracle schemas use reference-SQL-selected tables and views with compact source DDL, hiding SQL. This tests relevant schemas, changing coverage and input length; it does not test retrieval.

[3]https://github.com/UKPLab/emnlp2026-core-t

Schema2Bench's pooled TEF advantage over Template persists in both Base conditions; the Ring contrast crosses zero with oracle schemas. Neutron's oracle TEF and gap favor Ring. Reference unknowns number 660 and 663 out of 1,940 in the two conditions. Unresolved gaps and wide ranking and regret bounds do not establish better target model selection.

a) Seven-database diagnostic: A separate ten-model diagnostic on the seven earlier databases finds no general Schema2Bench advantage. Equal-domain means of Verified direct TEF bounds are $[-0.1055, +0.0750]$, $[+0.0206, +0.2190]$, $[-0.5911, +0.1033]$ and $[-0.4155, +0.4507]$ for S2B minus Template, Ring, Omni and Synth fallback, respectively. These average domain contrasts rather than pooling tasks. S2B has lower TEF than Ring on TPC-H, higher TEF on Chinook, Instacart, SFDFFD and TPC-DS, and unresolved differences on IMDb and USAccidents.

Task counts vary: Omni uses an 8,192-token limit on TPC-DS and partial outputs on TPC-H; Synth uses fallback generation. Independent target references are unverified, precluding gap, ranking and regret claims.

F. Discussion

Schema2Bench is supported for source-preserving migration: it improves both objectives over Template and Verified TEF over the three external adaptations. Ring's lower gap and mean regret favor it for target model selection, despite fewer Astra-

TABLE XI

EXTERNAL PILOTS: S2B MINUS COMPARATOR; NEGATIVE FAVORS S2B. TACO USES 750 TASKS PER METHOD AND 595 REFERENCES WITH DATABASE AND REFERENCE CHECKS; BEAVER USES 472 AND ALL 194, RESPECTIVELY. BOTH USE MATCHING SOURCE CONTEXT, POOLED TEF AND DOMAIN-AVERAGED GAP; ROWS SEPARATE SCORING AND SCHEMAS.

Setting	Comparator	ΔTEF	ΔG
TACO: retrieved schema	Template	$[-0.006024, +0.016398]$	$[-0.030517, +0.044555]$
TACO: retrieved schema	Ring	$[+0.133673, +0.280760]$	$[+0.148866, +0.277121]$
BEAVER: retrieved, Base	Template	$[-0.146373, -0.033660]$	$[-0.081035, +0.060364]$
BEAVER: retrieved, Base	Ring	$[-0.242054, -0.013075]$	$[-0.056789, +0.099330]$
BEAVER: oracle schema, Base	Template	$[-0.142702, -0.028519]$	$[-0.083884, +0.049262]$
BEAVER: oracle schema, Base	Ring	$[-0.130602, +0.168975]$	$[-0.051570, +0.064633]$
BEAVER: retrieved, Verified	Template	$[-0.146163, -0.037765]$	$[-0.081035, +0.060364]$
BEAVER: retrieved, Verified	Ring	$[-0.235815, -0.016203]$	$[-0.056789, +0.099330]$

supported pairs (Table VIII). The preferred method therefore depends on the measurement objective.

Retrospective tolerance choices do not establish non-inferiority, which requires prospective margins and sufficient precision.

Most improvement precedes selection, whose contribution is bounded adjustment and coverage diagnosis. Screening, budgets and shared data restrict generalization. External results are mixed; semantic-risk bounds fix unflagged scores.

Table VII includes failed-call tokens. Artifacts support offline score, feasibility and execution-evidence checks. Full reproduction needs external models, data and environments; redistribution rights remain unverified.

VI. RELATED WORK

A. Text-to-SQL

Text-to-SQL research improves schema linking, representation learning and constrained decoding [29–33]. DAIL-SQL studies prompting and example selection; CodeS develops specialized open models [1, 2]. PURPLE retrieves demonstrations by operator composition; KaSLA models schema linking as knapsack optimization [34, 35]. Long-context NL2SQL studies how schema, values and context affect accuracy and latency [36]. These methods improve predictors; Schema2Bench constructs workloads to compare them on new databases.

DIN-SQL, MAC-SQL and SHARE use decomposition, collaboration or correction [37–39]; CycleSQL uses database-grounded explanations as feedback [3]. These checks improve predictions but do not establish benchmark fidelity. Execution agreement itself requires qualification through test suites, formal checks and reference audits [9, 40–42].

Spider, BIRD and conversational benchmarks provide shared evaluation tasks [4, 5, 43, 44]. Spider 2.0, KaggleDBQA, ScienceBenchmark, BEAVER and customer-log evaluations extend the range of database demands [28, 45–48]. Perturbation benchmarks isolate linguistic, knowledge, schema and query changes [11, 49–53]. SQLMorph adds join expansion and fine-grained execution metrics; VLD-Bench studies discrepancies between question values and database entries [8, 10].

These diagnostics isolate failures under task changes, without establishing source fidelity or target alignment after migration. Following evidence-centered design and concurrence research [12, 13, 54], we separately assess source response profiles and target accuracy gaps, rankings and regret, retaining uncertainty when execution or semantic evidence is insufficient.

B. Data Synthesis

Synthetic question–SQL pairs support pretraining and adaptation [55–57]. RingSQL instantiates schema-independent templates; OmniSQL synthesizes large-scale training data [6, 7]. SING-SQL, EvolSQL and SynSQL study other synthesis settings [58–60]. These training objectives do not establish evaluation fidelity. We compare local adaptations at fixed workload sizes and isolate wording effects with SQL fixed.

Dataset distillation approximates learning from larger datasets with small synthetic training sets [61, 62]; gradient-matching condensation matches training gradients. Schema2Bench instead constructs cross-database test workloads without updating evaluators. Distilled data are assessed through learned models; migrated benchmarks require validation of the model comparisons they support.

Data selection reduces training or evaluation cost [63–65]. TinyBenchmarks estimates benchmark performance from selected items; SMART filters items; SubLIME and FAFI study rank preservation [14, 66–68]. Schema2Bench uses moment matching and benefit–cost exchanges [69–71] to control SQL composition under domain quotas. Migration also changes grounding and candidate populations. We bound fidelity changes from structural exchanges and diagnose shortages that selection cannot fix. Small additional selection gains and mixed external results delimit the benefits.

VII. CONCLUSION

Schema2Bench combines Benchmark Construction, Workload Selection and Benchmark Evaluation for source-preserving migration. On eight BIRD databases, it improves both source fidelity and target gap over Template, and achieves lower TEF than three external adaptations under Verified scoring and declared weights. Most gain precedes selection; coverage bounds identify deficits selection cannot repair. Lower external target gaps, semantic uncertainty and mixed external-domain results delimit this advantage.

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We used OpenAI Codex to assist with language polishing, grammar correction, and improving the readability of this manuscript. The authors are responsible for the scientific content and conclusions.

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